

SG-319-250

 Precision Industries Testing Department	GENERATING-SET				Model No: PI 250 C		60 Hz	
	LOAD TEST				Type: Close		Job No: 9428	
					Color: White		Date: 12/08/2018	

Serial Number:								
PI Model No:	PI 250 C	Prime Power:	225	[kVA]	P.F (Load):	1	Voltage:	220

Generating Set Details:

Engine:	Cummins	Model No: QSB7 - G5	Serial No: 84276140
Alternator:	Stamford	Model No: UCI 274 H1	Serial No: N17D166381
Controller:	DSE	Model No: 6120	Serial No: 6754517
C. Breaker:	Metasol 800 A	Model No: ABN 803 C	Serial No: N/A
C. Transformer:	Ziegler 800/5A	Model No: N/A	Serial No: N/A
AVR:	Stamford	Model No: SX 460	Serial No: 111505 - 3959
Cooling System: Water Cool	Enclosure Type: Canopy	Protection Class: IP23	

Load Test Ambient Conditions:

Temperature	38	[°C]	Altitude:	0	[m]	Humidity:	53	%
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Generating Set Rating:

Testing Full Load Current:	472.38	Amps	Nominal Current:	590.5	Amps
Power Correction Factor (De-rating):	0.000	%	Corrected Prime Power:	0.0	[kVA]

LOAD TEST

Load	Run Time	Freq	Speed	VOLTAGE L-L			CURRENT			Power	Coolant Temp	Oil Pressure	Charg-Alt.
				L1	L2	L3	L1	L2	L3				
%	min	Hz	RPM	V	V	V	A	A	A	kW	C	bar	VDC
0 %	5	60	1800	220	220	220	-	-	-	-	70	4.7	14.4
25 %	5	60	1800	220	220	220	118	117	116	45.0	81	4.62	14.4
50 %	5	60	1800	220	220	220	239	238	238	91.0	83	4.00	14.4
75 %	5	60	1800	220	220	221	356	355	356	135.0	85	3.92	14.6
100 %	10	60	1800	220	221	221	474	474	473	181.0	89	3.48	14.8
110 %	10	60	1800	221	221	221	519	519	518	198.0	94	3.32	14.8

LOAD ACCEPTANCE TEST

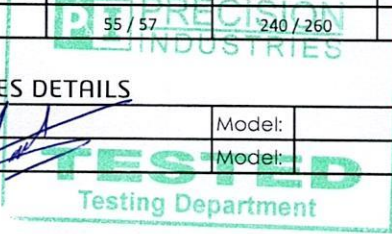
Suddenly Block Load %	Voltage [V]	Frequency [Hz]	Current [A]	Response Time [sec]
50%	220	60.0	239	0
0%	227 - 220	61.7 - 60.0	0	2
50%	216 - 220	58.3 - 60.0	242	2

PROTECTION SYSTEM TEST

Over Frequency Warning / SD	Under Frequency Warning / SD	Over Voltage Warning / SD	Under Voltage Warning / SD	Oil Pressure SD	Coolant Level Warning / SD	Coolant Temp Warning / SD
65 / 67	55 / 57	240 / 260	190 / 180	1.0	OK	103 / 108 C

ACCESSORIES DETAILS

Model:		Serial No.	
Model:		Serial No.	



PI Document no.	PI-TST-LT-01	Rev.	01
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Test Certificate

FOR
SALIENT POLE, SELF EXCITED / PMG EXCITED AND SELF REGULATED AC GENERATOR

Machine No. **N17D166381**

Type Test has been conducted on similar design machine
STANDARDS: Generator generally conforms to BS 5000 : Part 3,
Tested in accordance with BS EN 60034-1/IS/IEC 60034-1
Rotor dynamically balanced to BS ISO 1940 - 1

ROUTINE TEST CERTIFICATE FOR BRUSHLESS AC GENERATOR:

FRAME: UCI274H1	AVR: SX460	AVR NO: 1115053959	ENCLOSURE: IP23	
RATED KVA: 200	RATED VOLTS: 400	RATED AMPS: 288.7	FREQUENCY: 50	PHASE: 3
POWER FACTOR: .8	STR CONN: Series Star	STR WINDING: 311	DUTY: Base Continuous	WIRE: 4-Wire
EXC. VOLTS: 58	EXC.AMPS: 2.3	AMB TEMP: 40		

FOR 3 PHASE SEQUENCE LOOKING FROM DRIVE END FOR CLOCKWISE ROTATION :U-V-W

WINDING	COLD RESISTANCE IN Ohm(+/- 10 Milli Ohm) AT 20°C (MAX VALUES)	INSULATION RESISTANCE (M- Ohm)	HV TEST 2KV FOR 1 MIN	INSULATION CLASS
STATOR	0.03371	2+	OK	H
ROTOR	1.9663	2+	OK	H

REGULATION TEST

FREQ.	VOLTS	AMPS	KVA	PF	KW	EXC.VOLT	EXC.AMP	% LOAD
52	401.35	0.00	0.00	0.00	0.00	12	.5	0.00
50	400	288.7	200	.8	160	58	2.3	100.00

REGULATION = (N.L.VOLTAGE-F.L.VOLTAGE / RATED VOLTAGE)*100 = 0.34 %

Item Description: AC Generator.UC27.4.H.1.311.,3.,,11.5.,SX460.Black - RAL9011
Remarks:

Date: 23-APR-17	Job No: 3481896-010	AUTHORISED BY Manoj K Upadhyaya Quality Assurance
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Certificate Of Conformity

Quality System Approval
ISO 9001/2008

Frame Size : **UCI274H1**

Serial Number : **N17D166381**

Manufactured Date : 23-APR-17

KVA: **200**

Authorized By Quality Assurance: **Manoj K Upadhyaya**

EU DECLARATION OF CONFORMITY



**Generator
Technologies**

This synchronous A.C. generator is designed for incorporation into an electricity generating-set and fulfils all the relevant provisions of the following EU Directive(s) when installed in accordance with the installation instructions contained in the product documentation:

2014/35/EU Low Voltage Directive
2014/30/EU The Electromagnetic Compatibility (EMC) Directive

and that the standards and/or technical specifications referenced below have been applied:

EN 61000-6-2:2005	Electromagnetic compatibility (EMC). Generic standards - Part 6-2: Immunity for industrial environments
EN 61000-6-4:2007+A1:2011	Electromagnetic compatibility (EMC). Generic standards - Part 6-4: Emission standard for industrial environments
EN ISO 12100:2010	Safety of machinery - General principles for design - Risk assessment and risk reduction
EN 60034-1:2010	Rotating electrical machines - Part 1: Rating and performance
BS ISO 8528-3:2005	Reciprocating internal combustion engine driven alternating current generating sets - Part 3: Alternating current generators for generating sets
BS 5000-3:2006	Rotating electrical machines of particular types or for particular applications - Part 3: Generators to be driven by reciprocating internal combustion engines - Requirements for resistance to vibration

This declaration has been issued under the sole responsibility of the manufacturer. The object of this Declaration is in conformity with the relevant Union harmonization Legislation.

The name and address of authorised representative, authorised to compile the relevant technical documentation, is the Company Secretary, Cummins Generator Technologies Limited, 49/51 Gresham Road, Staines, Middlesex, TW18 2BD, U.K.

Signed:

Date: 01st February 2016

Name, Title and Address:

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PE2 6FZ

Description UC1274H1

Serial Number N17D166381

450-16381-F